

Groups and Lie Algebras: Exercise Sheet 10 — Solutions

1 The Killing form

Throughout, $\kappa(X, Y) = \text{tr}(\text{ad}_X \text{ad}_Y)$, where $\text{ad}_X(Y) = [X, Y]$.

(i) *The Killing form of $\mathfrak{sl}(2)$.* In the basis H, E, F the brackets are

$$[H, E] = 2E, \quad [H, F] = -2F, \quad [E, F] = H.$$

In the ordered basis (H, E, F) the adjoint matrices are

$$\text{ad}_H = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -2 \end{pmatrix}, \quad \text{ad}_E = \begin{pmatrix} 0 & 0 & 1 \\ -2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \text{ad}_F = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}.$$

Taking traces of products gives the nonzero entries

$$\kappa(H, H) = \text{tr}(\text{ad}_H^2) = 0^2 + 2^2 + (-2)^2 = 8, \quad \kappa(E, F) = \text{tr}(\text{ad}_E \text{ad}_F) = 4,$$

while $\kappa(H, E) = \kappa(H, F) = \kappa(E, E) = \kappa(F, F) = 0$. Hence, in the basis (H, E, F) ,

$$\kappa = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 4 & 0 \end{pmatrix}.$$

This is nondegenerate (so $\mathfrak{sl}(2)$ is semisimple, as it must be), and one checks the familiar relation $\kappa(X, Y) = 4 \text{tr}(XY)$ in the fundamental representation.

(ii) *Invariance* $\kappa([X, Y], Z) = \kappa(X, [Y, Z])$. Since $\text{ad}: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ is a Lie-algebra homomorphism, $\text{ad}_{[X, Y]} = [\text{ad}_X, \text{ad}_Y]$. Using this and cyclicity of the trace,

$$\begin{aligned} \kappa([X, Y], Z) &= \text{tr}([\text{ad}_X, \text{ad}_Y] \text{ad}_Z) = \text{tr}(\text{ad}_X \text{ad}_Y \text{ad}_Z) - \text{tr}(\text{ad}_Y \text{ad}_X \text{ad}_Z), \\ \kappa(X, [Y, Z]) &= \text{tr}(\text{ad}_X [\text{ad}_Y, \text{ad}_Z]) = \text{tr}(\text{ad}_X \text{ad}_Y \text{ad}_Z) - \text{tr}(\text{ad}_X \text{ad}_Z \text{ad}_Y). \end{aligned}$$

By cyclicity $\text{tr}(\text{ad}_Y \text{ad}_X \text{ad}_Z) = \text{tr}(\text{ad}_X \text{ad}_Z \text{ad}_Y)$, so the two right-hand sides coincide. $\square$

(iii) *Invariance under automorphisms of $\mathfrak{g}$.* Let $\phi: \mathfrak{g} \rightarrow \mathfrak{g}$ be a Lie-algebra automorphism. For all X, Y ,

$$\text{ad}_{\phi(X)}(Y) = [\phi(X), Y] = \phi[X, \phi^{-1}(Y)] = (\phi \text{ad}_X \phi^{-1})(Y),$$

i.e. $\text{ad}_{\phi(X)} = \phi \text{ad}_X \phi^{-1}$. Therefore

$$\kappa(\phi X, \phi Y) = \text{tr}(\phi \text{ad}_X \phi^{-1} \phi \text{ad}_Y \phi^{-1}) = \text{tr}(\phi \text{ad}_X \text{ad}_Y \phi^{-1}) = \text{tr}(\text{ad}_X \text{ad}_Y) = \kappa(X, Y).$$

In particular, for $g \in G$ the map Ad_g (the differential of conjugation by g) is such an automorphism, so $\kappa(\text{Ad}_g X, \text{Ad}_g Y) = \kappa(X, Y)$. This is exactly the fact used in 3(d). $\square$

(iv) *Structure-constant formula.* Write $[T^a, T^b] = \sum_c f^{ab}_c T^c$. The matrix of ad_{T^a} in the basis $\{T^c\}$ has entries $(\text{ad}_{T^a})^c_b = f^{ab}_c$ (row c , column b). Hence

$$\kappa(T^a, T^b) = \text{tr}(\text{ad}_{T^a} \text{ad}_{T^b}) = \sum_{c,e} (\text{ad}_{T^a})^c_e (\text{ad}_{T^b})^e_c = \sum_{c,e} f^{ae}_c f^{be}_c = \sum_{c,e} f^{bc}_e f^{ae}_c,$$

the last equality being a relabelling. This is the claimed identity; note it is manifestly symmetric under $a \leftrightarrow b$. $\square$

2 The root diagram and Weyl group of G_2

The G_2 Dynkin diagram has two nodes joined by a triple bond, the arrow pointing from the long simple root α_1 to the short simple root α_2 . The corresponding Cartan integers are

$$\langle \alpha_2, \alpha_1^\vee \rangle = -1, \quad \langle \alpha_1, \alpha_2^\vee \rangle = -3,$$

which fixes the angle between the simple roots at 150° and the length ratio $|\alpha_1|/|\alpha_2| = \sqrt{3}$.

Simple reflections (defined by their action on the simple roots).

$$\begin{aligned} s_1(\alpha_1) &= -\alpha_1, & s_1(\alpha_2) &= \alpha_2 + \alpha_1, \\ s_2(\alpha_2) &= -\alpha_2, & s_2(\alpha_1) &= \alpha_1 + 3\alpha_2. \end{aligned}$$

Writing a general element as $m\alpha_1 + n\alpha_2 \equiv (m, n)$, linearity gives

$$s_1: (m, n) \mapsto (n - m, n), \quad s_2: (m, n) \mapsto (m, 3m - n).$$

The Weyl group. Let $r = s_1 s_2$. Then $r(m, n) = s_1(m, 3m - n) = (2m - n, 3m - n)$, and iterating on $\alpha_1 = (1, 0)$:

$$(1, 0) \rightarrow (2, 3) \rightarrow (1, 3) \rightarrow (-1, 0) \rightarrow (-2, -3) \rightarrow (-1, -3) \rightarrow (1, 0),$$

so r has order 6. Together with $s_1^2 = s_2^2 = \text{id}$ this gives

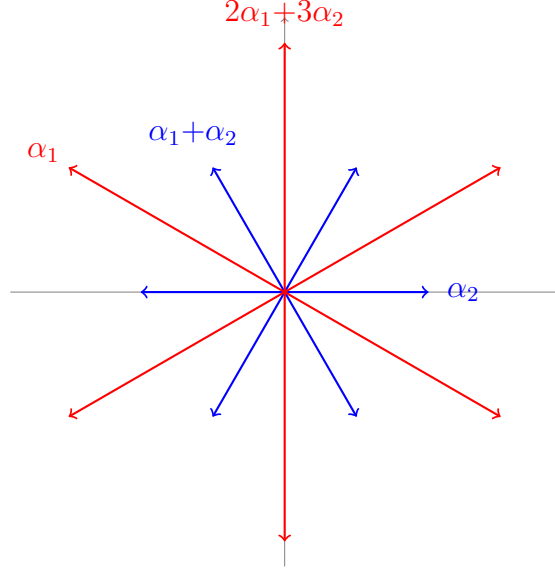
$$W(G_2) = \langle s_1, s_2 \mid s_1^2 = s_2^2 = (s_1 s_2)^6 = \text{id} \rangle \cong D_6, \quad |W| = 12,$$

the dihedral group of order 12 (symmetries of a regular hexagon).

The root system. Applying W to the simple roots produces 12 roots, the six positive ones being

$$\underbrace{\alpha_2, \alpha_1 + \alpha_2, \alpha_1 + 2\alpha_2}_{\text{short}}, \quad \underbrace{\alpha_1, \alpha_1 + 3\alpha_2, 2\alpha_1 + 3\alpha_2}_{\text{long}},$$

together with their negatives. Geometrically the short roots point at angles $0^\circ, 60^\circ, \dots$ and the long roots at $30^\circ, 90^\circ, \dots$, i.e. the roots sit at 30° intervals, alternating short/long:



(blue = short, red = long). $\square$

3 The YM action functional

a) Background. The gauge field is a $\mathfrak{g}$ -valued 1-form $A = \sum_i A_i dx^i$, $A_i: \mathbb{R}^4 \rightarrow \mathfrak{g}$. The first term $gA_i g^{-1} = \text{Ad}_g(A_i)$ is the adjoint action; the second term $(\partial_i g)g^{-1} \in \mathfrak{g}$ because $\partial_i g \in T_g G$ and right translation by g^{-1} carries it to $T_e G = \mathfrak{g}$. The Killing form κ is symmetric, invariant (1.ii) and Ad-invariant (1.iii); these are the only facts about κ we use below.

b) The case $G = SU(2)$. Take the anti-Hermitian basis $T^a = -\frac{i}{2}\sigma^a$ of $\mathfrak{su}(2)$, so that

$$[T^a, T^b] = \epsilon^{abc} T^c, \quad \kappa(T^a, T^b) = -2\delta^{ab}.$$

Writing $A_i = \sum_a A_i^a T^a$ with real components $A_i^a: \mathbb{R}^4 \rightarrow \mathbb{R}$,

$$F_{ij}^a = \partial_i A_j^a - \partial_j A_i^a + \epsilon^{abc} A_i^b A_j^c.$$

Since $\kappa(F_{ij}, F_{ij}) = -2 \sum_a (F_{ij}^a)^2$, the action becomes

$$S_{\text{YM}}[A] = \frac{1}{2e^2} \sum_{i,j} \int_{\mathbb{R}^4} \sum_a (F_{ij}^a)^2 d^4x \geq 0,$$

the positivity coming precisely from the minus sign together with the negative-definiteness of κ on the compact form $\mathfrak{su}(2)$. Infinitesimally, $g = e^\lambda$ with $\lambda = \lambda^a T^a$ gives

$$\delta A_i = [\lambda, A_i] - \partial_i \lambda, \quad \delta A_i^a = -\partial_i \lambda^a + \epsilon^{abc} \lambda^b A_i^c.$$

c) *Transformation of F .* Regard A_i as the connection operator $D_i = \partial_i + A_i$ (acting on $\mathfrak{g}$ -valued fields). A direct check using $g \partial_i (g^{-1}) = -(\partial_i g) g^{-1}$ shows

$$g (\partial_i + A_i) g^{-1} = \partial_i + (g A_i g^{-1} - (\partial_i g) g^{-1}) = \partial_i + A_i^g.$$

Moreover one computes, as operators,

$$[D_i, D_j] = [\partial_i + A_i, \partial_j + A_j] = \partial_i A_j - \partial_j A_i + [A_i, A_j] = F_{ij},$$

the second-derivative terms cancelling. Hence

$$F_{ij}^g = [g D_i g^{-1}, g D_j g^{-1}] = g [D_i, D_j] g^{-1} = g F_{ij} g^{-1} = \text{Ad}_g(F_{ij}),$$

so F transforms *homogeneously* in the adjoint representation; the inhomogeneous (derivative-of- g) pieces all cancel. $\square$

d) *Gauge invariance of the action.* Pointwise, using (c) and the Ad-invariance of κ established in 1(iii),

$$\kappa(F_{ij}^g, F_{ij}^g) = \kappa(\text{Ad}_g F_{ij}, \text{Ad}_g F_{ij}) = \kappa(F_{ij}, F_{ij}).$$

Since the integrand is unchanged at every point of $\mathbb{R}^4$,

$$S_{\text{YM}}[A^g] = -\frac{1}{4e^2} \sum_{i,j} \int_{\mathbb{R}^4} \kappa(F_{ij}^g, F_{ij}^g) d^4x = S_{\text{YM}}[A]. \quad \square$$